INTRODUCTION TO DATA ANALYTICS · FRANKFURT SCHOOL

AIRPORT DISRUPTION IN GERMANY

Forecasting daily ATFM arrival delays at 12 German airports, 2014–2025

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Terminal 1, Frankfurt. Orange on the board means delayed.

BEFORE WE TAKE OFF

YOUR FLIGHT IS LATE AGAIN. WHO DO YOU BLAME?

A The weather

B Air traffic control

C The airport itself

D *Ich weiß es nicht*

Keep your answer. We check it on slide 10.



OUR STORY · FRANKFURT AM MAIN

WE LEARNED „VERSPÄTUNG“ BEFORE „DANKE“

1 We arrived at FRA as new international students. The board was orange.

2 Long waits, missed connections, and the same announcement every time.

3 As data science students at Frankfurt School, we asked: could the data have seen it coming?

„Wir bitten um Ihr Verständnis.“



ZEIT IST GELD · THE BUSINESS CASE

7.8 YEARS

of waiting. That is how much ATFM arrival delay piled up at just 12 German airports between 2014 and 2025. Every minute of it idles crews, gates and passengers.

4.09 million
delay minutes in total

55%
of them at Frankfurt alone

1 in 16
airport-days is disrupted

THE PROJECT

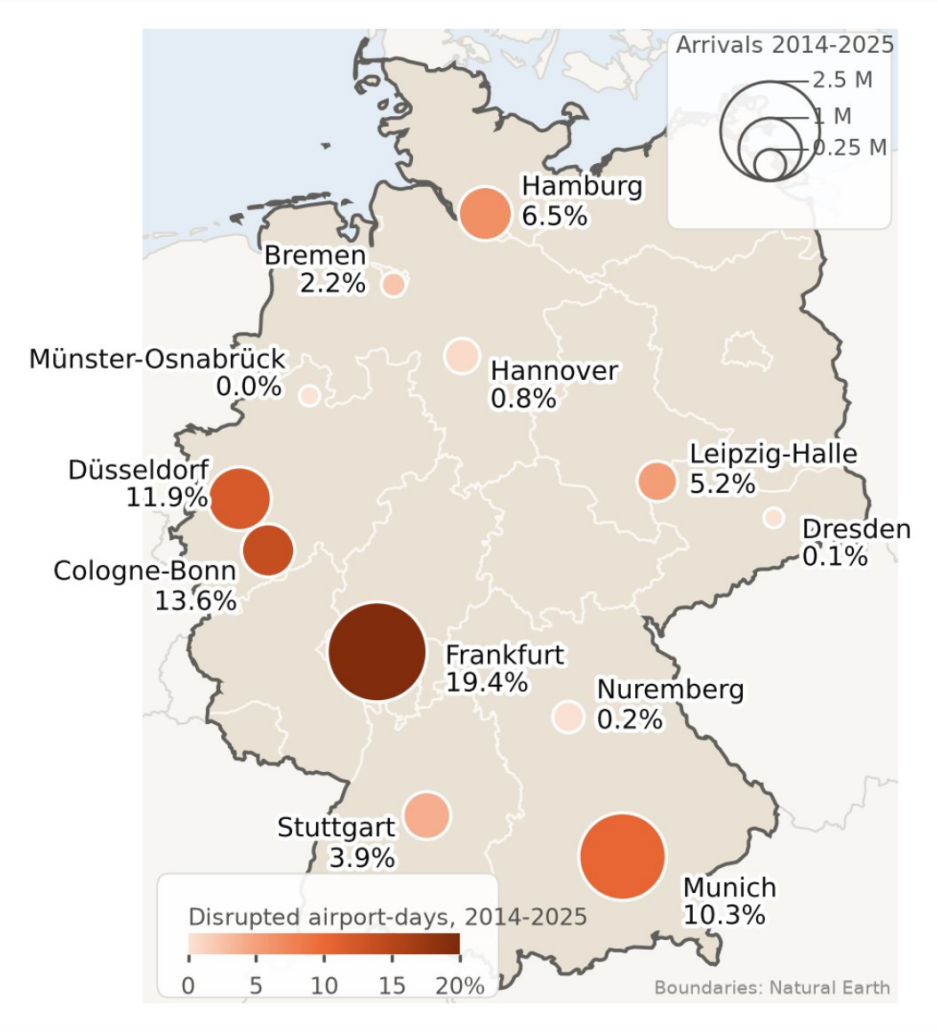
CAN PUBLIC DATA PREDICT A DISRUPTED DAY?

A day counts as **disrupted** when at least 2% of its arrivals get more than 15 minutes of ATFM delay. That is 6.2% of all days, and 19.4% at Frankfurt.

RISK Will today be disrupted? yes or no

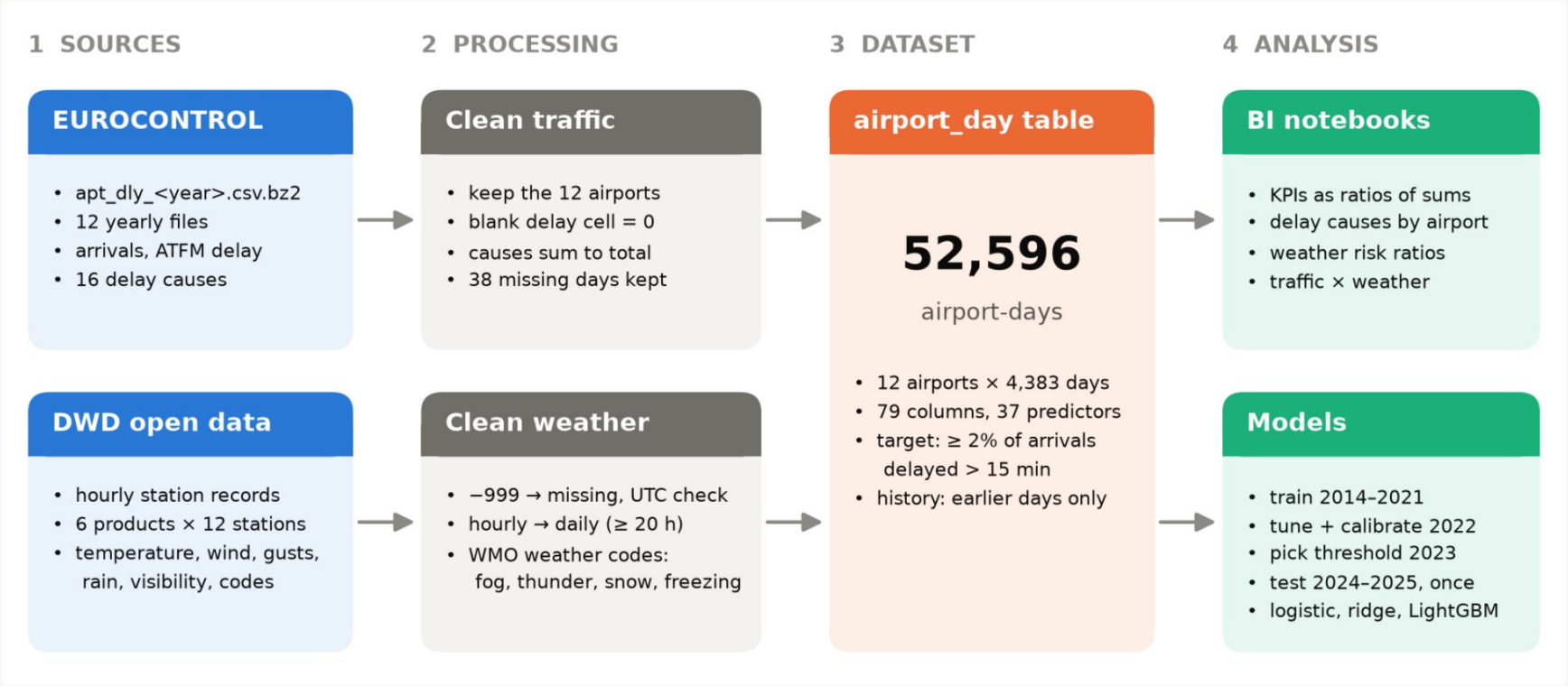
SEVERITY How many delay minutes per arrival?

12 airports · 2014–2025 · 52,596 airport-days · EUROCONTROL and DWD open data



DATA SOURCES

TWO PUBLIC SOURCES BECOME ONE TABLE



84 raw files, each logged with a SHA-256 hash · a rebuild from raw gives an identical dataset · the 2025 file even had Umlaut trouble: two text encodings in one file

FRANKFURT · NOVEMBER 2024

ONE ROW PER AIRPORT PER DAY, 79 COLUMNS

Date	Arrivals	Delay, min	Over 15 min	Share	Max gust	Disrupted
17 Nov	588	0	0	0.0%	13.1 m/s	no
18 Nov	597	0	0	0.0%	11.9 m/s	no
19 Nov	522	19,266	293	56.1%	20.1 m/s	yes
20 Nov	593	276	4	0.7%	13.3 m/s	no
21 Nov	588	0	0	0.0%	12.4 m/s	no
22 Nov	589	1,160	31	5.3%	14.1 m/s	yes

19 November 2024 was the worst day of the test years: more than half of all arrivals were delayed, coded as aerodrome capacity.

Share = arrivals delayed over 15 minutes ÷ arrivals · a day is disrupted from 2% · 16 delay causes, 20 weather and 10 history features per row

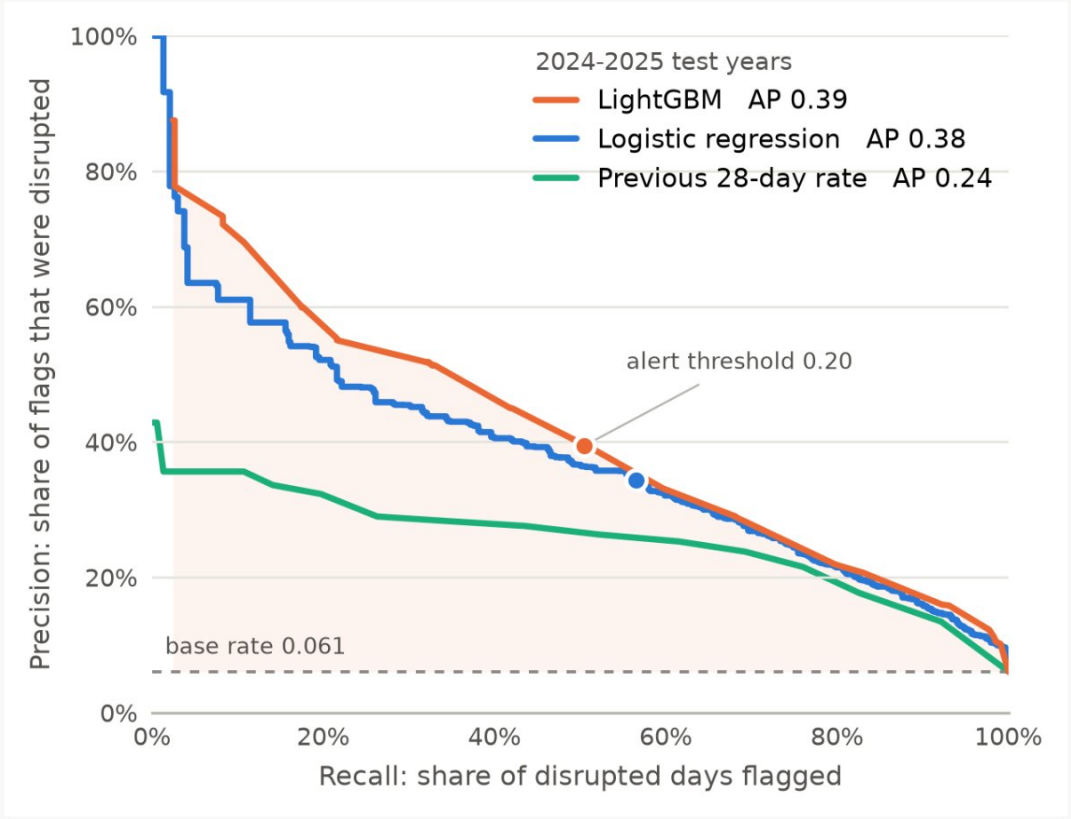
MODEL CHOICE

TWO LINEAR MODELS, TWO TREE MODELS, ONE HONEST SPLIT

	SIMPLE AND LINEAR	GRADIENT-BOOSTED TREES
RISK disrupted, yes or no	**Logistic regression** One readable weight per input. Fast and easy to explain.	**LightGBM classifier** Learns thresholds and combinations, like storms that only hurt busy hubs.
SEVERITY delay minutes per arrival	**Ridge regression** Linear with a penalty, so overlapping history features stay calm.	**LightGBM, Tweedie** Made for targets that are mostly zero with a long tail.

Train 2014–2021	Tune + calibrate 2022	Pick threshold 2023	Test 2024–25, scored once

Split by time, never shuffled. Each model has to beat a simple rule: the last 28 days.



RESULTS · TEST YEARS 2024–25

LIGHTGBM BEATS THE 28-DAY RULE

Average precision 0.39 against 0.24, on 8,771 airport-days scored once.

50% of disrupted days caught at the 0.20 alert threshold

0.63 AP at Frankfurt, against 0.30 for the 28-day rule

3 / 52 disrupted days caught at Stuttgart. Still homework.

Ehrlich gesagt: 89% of days have no delay, so "nothing happens" is hard to beat. Our delay-minutes model won by only 0.007 MAE.

Settings fixed on 2022–2023 before the test years · observed weather stands in for a forecast

RESULTS · ALL MODELS, TEST YEARS 2024–25

THE SCOREBOARD

DISRUPTED DAY? · CLASSIFIERS

	AP	ROC-AUC	Brier	Prec.	Recall
LightGBM	**0.389**	**0.895**	**0.045**	**0.39**	0.50
Logistic regression	0.376	0.889	0.046	0.34	0.56
28-day rate	0.242	0.848	0.050	0.25	**0.62**
Constant rate	0.061	0.500	0.057	–	–

DELAY MIN PER ARRIVAL · REGRESSORS

	MAE	RMSE	R^2
LightGBM (Tweedie)	**0.200**	**0.947**	**0.16**
Ridge regression	0.290	0.957	0.14
Always zero	0.207	1.054	-0.04
28-day average	0.298	0.989	0.08

8,771 airport-days scored once · bold: best per column · grey rows: baselines · lower is better for Brier, MAE, RMSE · precision and recall at the 0.20 alert threshold chosen on 2023

BI · NETWORK KPIS, 2014–2025

WEATHER CAUSES TWO THIRDS OF ALL DELAY MINUTES

TOTAL ARRIVALS	DELAY MIN PER 1,000 ARRIVALS	DISRUPTION RATE	WEATHER SHARE OF DELAY
8.93 M	**458**	**6.2%**	**67.3%**
How much traffic are we analysing?	How severe is delay relative to traffic?	How often do airport disruptions occur?	How much ATFM delay is weather-related?

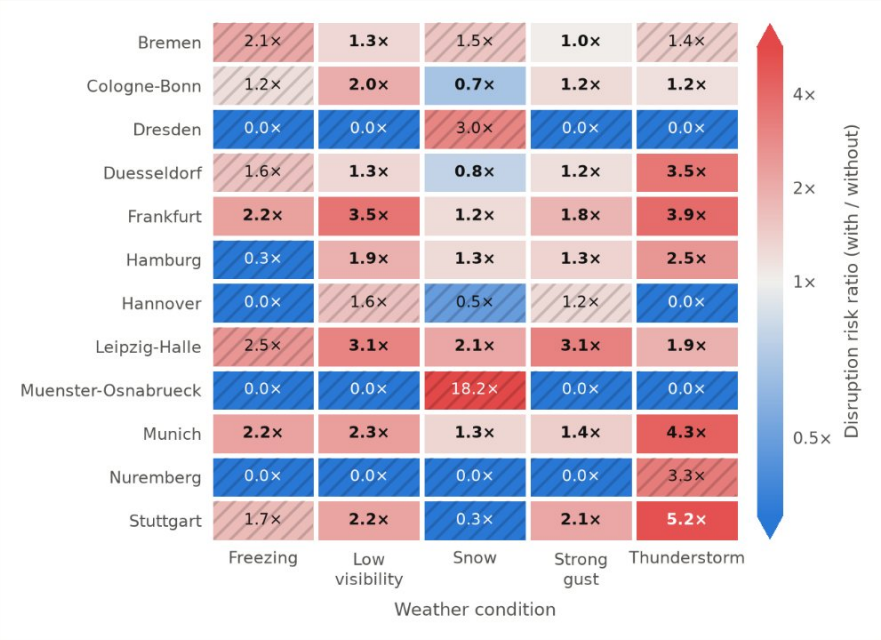
Slide 2, solved: the answer is mostly **A, the weather**. Sorry, air traffic control.

Ratios of sums over all 52,596 airport-days, so each airport counts by its traffic

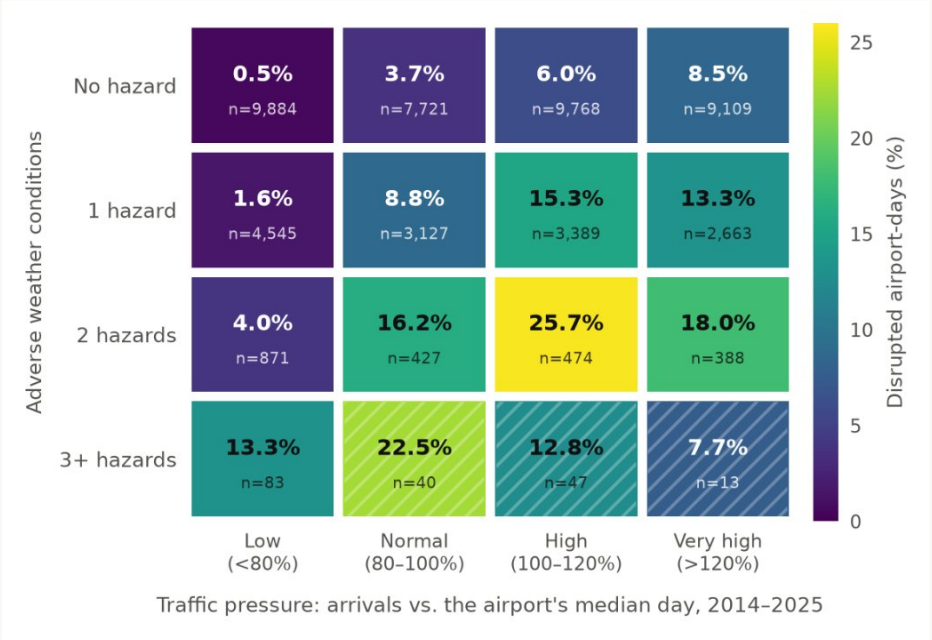
BI · WHERE AND WHEN IT HITS

STORMS AND BUSY DAYS DRIVE DISRUPTION

Thunderstorms make a day 3.9 to 5.2 times as likely to be disrupted at Frankfurt, Munich and Stuttgart.



A busy day with two hazards is disrupted 25.7% of the time. A quiet, calm day: 0.5%.



CREW

WHO DID WHAT

CREW 01

Siddharth Jain

- Data engineering: sources, pipeline, airport-day table
- LightGBM classifier and LightGBM regressor

CREW 02

Manasvi Japi

- Logistic regression and Ridge regression
- Presentation

CREW 03

Rong Qin

- BI analysis and KPIs
- Micropublication
- Presentation

Code, data and paper: github.com/sidnov6/german-airport-disruption

LANDED · ON TIME, FOR ONCE

DANKE & PROST!

Thank you and cheers. Please stay seated until the Q&A sign is switched off.

FRAGEN? github.com/sidnov6/german-airport-disruption